



Experimental and numerical investigation on the bubble breakup and coalescence characteristics in the highly efficient static mixers

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ARTICLE INFO

Keywords:

Static mixer
Coalescence and breakup models
Bubble size distribution
CFD-PBM coupled model
Residence time distribution

ABSTRACT

CFD-PBM and experiment methods were applied to study the bubble dispersion characteristics in the static mixers with three twisted blades (TKSM). The gas-liquid velocity ratio was selected as drag correction factors. The simulation data was compared with experimental values at different superficial gas velocity and the results revealed that the modified model could effectively enhance the calculation accuracy of gas holdup. Furthermore, bubble size distribution (BSD) was well predicted by Luo-Prince model and the mean relative error (MRE) of axial d_{32} between the prediction of Luo-Prince model and experimental value was 2.86 %. Meanwhile, the influences of bubble collision mechanisms and coalescence correctors on the numerical prediction were quantitatively analyzed. The effects of turbulent fluctuation, viscous shear, wake entrainment and buoyancy driven were considered for the prediction of bubble collision frequency in TKSM. The prediction precision of cumulative probability distribution (CPD) was promoted by 17.26 % when the correctors obtained from Energy-Minimization Multi-Scale (EMMS) method were used. In addition, empirical parameter C_{Das} was evaluated and the results showed that the Das coalescence model could accurately describe the BSD when C_{Das} was 0.01–0.1. The backmixing performance of the gas phase in the TKSM exceeds that of the liquid phase, which is distinct from that in vortex unit.

1. Introduction

The study of multiphase flow is of great significance for industrial production which involves processes such as heat and mass transfer, phase transition and so on. The comprehensive information of these processes can be well obtained by numerical simulation. In the simulation process of gas-liquid system, the simulation accuracy is largely determined by the interfacial forces. One of the most important forces among them is drag force. During the past decades, plenty of work had been done on the development of drag force model (Tas-Koehler et al., 2021). The research of drag force mainly focused on two aspects. Firstly, Schiller & Naumann model as the most famous single bubble drag force model had been extensively applied in two-phase flow systems (Schiller and Naumann, 1933; Haddadi et al., 2020a; Guo et al., 2021; Meng et al., 2021). Single bubble drag coefficient was proposed considering bubbles shape (Ishii and Zuber, 1979; Tomiyama et al., 1998). In addition, the influence of turbulent characteristics on the flow feature was considered (Bakker and Van den Akker, 1994; Brucato et al., 1998;

Buffo et al., 2016). More information on single bubble drag modeling can refer to these works (Chuang and Hibiki, 2017; Khan et al., 2020; Tas-Koehler et al., 2021). Secondly, the influence of bubble swarms was considered in the drag force modeling. Roghair et al. (2013) used Direct Numerical Simulation and Lagrange method to obtain the drag force of the bubble clusters. Wang and Wang (2007) considered the wake acceleration effect of large bubbles. The Energy-Minimization Multi-Scale (EMMS) model was originally proposed and applied in gas-solid systems (Li et al., 1999; Ge and Li, 2002), and has become an effective multi-scale approach in gas-liquid systems (Zhang et al., 2023c). Yang et al. (2007) proposed dual-bubble-size (DBS) modeling on the basis of EMMS concept. Based on previous work (Wang and Wang, 2007), Yang et al. (2018) added the hindering influence of small bubbles on interfacial force. Yan et al. (2019) introduced a density correction term to make it suitable for high pressure system by using EMMS approach. In addition, the selections of bubble coalescence and breakup models are also considerable challenges for simulation results except for interfacial forces.

Many researchers have been dedicated to proposing mathematical

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<https://doi.org/10.1016/j.ces.2023.119692>

Received 4 August 2023; Received in revised form 5 December 2023; Accepted 27 December 2023

Available online 31 December 2023

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Nomenclature		cross-section [m]	
A	flow section area [m ²]	<i>Subscripts</i>	
a(v,v')	bubble coalescence rate [s ⁻¹]	c	continuous phase
B _{Ag}	Bubble birth rate due to coalescence	d	dispersed phase
B _{Br}	Bubble birth rate due to breakage	G	gas phase
b(v)	bubble breakup rate with volume v [s ⁻¹]	L	liquid phase
C _{D,0}	drag coefficient of single bubble	i	ith phase
C _D /C _{D,0}	drag correction factor of bubble swarms	<i>Greek Symbols</i>	
D _{Ag}	Bubble death rate due to coalescence	α _d	dispersed phase volume fraction
D _{Br}	Bubble death rate due to breakage	α _G	gas holdup
D _H	hydraulic diameter [m]	β(f _v ,d)	daughter bubble size distribution function
d _b	diameter of the bubble [mm]	μ	viscosity [Pa·s]
d ₃₂	Sauter mean diameter [mm]	μ _t	turbulent eddy viscosity [Pa·s]
Eo	Eötvös number	ε	turbulent dissipation rate [m ² ·s ⁻³]
F	Interfacial force [N]	ω	specific turbulence dissipation rate [s ⁻¹]
f	Darcy friction factor	ρ	density [kg·m ⁻³]
G _k	production rate of k due to turbulence [w·m ⁻³]	σ _ω	turbulence Prandtl number for ω
G _{kb}	production rate of k due to buoyancy [w·m ⁻³]	σ _k	turbulence Prandtl number for k
G _ω	production rate of ω due to turbulence [w·m ⁻³]	η	Kolmogorov microscale
G _{ωb}	production rate of ω due to buoyancy [w·m ⁻³]	ζ	stress tensors
g	gravitational acceleration vector [m·s ⁻²]	<i>Abbreviations</i>	
k	kinetic turbulence energy [m ² ·s ⁻²]	BSD	Bubble size distribution
l	length of a single mixing element [m]	BIT	Bubble induced turbulence
Mo	Morton number	CFD	Computational Fluid Dynamics
n(v)	bubbles number density function	CPD	Cumulative probability distribution
P	wetted perimeter [m]	EMMS	Energy-Minimization Multi-Scale
Re	Reynolds number	GCI	Grid convergence index
Re _B	relative Reynolds number	KSM	Kenics static mixer
S _k	source of k due to turbulence [w·m ⁻³]	MRE	Mean relative error
S _ω	source of ω due to turbulence [w·m ⁻³]	MRT	Mean residence time
U _G	superficial gas velocity [m·s ⁻¹]	PBM	Population balance model
U _L	superficial liquid velocity [m·s ⁻¹]	TKSM	Static mixers with three twisted leaves
V	volume of a bubble [mm ³]		
z	distance from the starting point of the mixing section to the		

models which could describe bubble breakup or coalescence phenomena. Luo and Svendsen (1996) believed that the bubble would break only when the energy contained in eddy exceeded the sum of the surface energy of the two daughter bubbles. Lehr et al. (2002) held the view that the breakup could happen only when the inertial force of the turbulent eddy exceeded the interfacial force of the smaller daughter bubbles. According to the aforementioned two viewpoints, Wang et al. (2003) thought that it was very difficult for a bubble to break into an infinitely small daughter bubble. Han et al., (2011, 2013, 2014) took into account the impact of turbulent eddies larger than the bubble size on the bubble breakup and the influence of surface oscillating frequency on the collision frequency. Then, multiple breakage concepts had been proposed to better calculate the breakup rate. Finally, the influence of energy spectrum function on bubble breakup rate was investigated. Xing et al. (2015) investigated the impact of the internal flow mechanism on the frequency of bubble breakup. They believed that bubbles were more prone to occur non-equal size breakup. Liao et al. (2015) also combined energy constraints with stress constraints. They considered that the bubble would break if the stress was greater than the maximum value of the critical stress dominated by energy or inertial force. Guo et al. (2016) deemed that bubbles could break in equal volume when the kinetic energy of turbulent eddy was dominated and bubbles tended to occur non-equal size breakup when the stress force was large enough. On the basis of the work already completed by Xing et al. (2015), Zhang et al. (2020a) considered the influence of viscous flow resistance on bubble breakup rate.

At present, the developed breakup model has been investigated in

bubble column (Shi et al., 2019), stirring tank (Wu et al., 2022a), fluidized bed (Wang et al., 2019), static mixer (Rabha et al., 2015), bioreactors (Sepehri and Sarrafzadeh, 2018; Sepehri and Sarrafzadeh, 2019; Sepehri et al., 2020) and other reactors. It is worth mentioning that the NAS approach which was proposed by (Sepehri and Sarrafzadeh, 2018; Sepehri and Sarrafzadeh, 2019; Sepehri et al., 2020) had good applicability and advance in the field of bioreactor. Compared with other devices, there are many advantages in static mixers, such as high mixing efficiency, low energy consumption and operation cost. They are widely applied in the fields of process intensification, nuclear energy safety and food processing, such as carbon dioxide absorption, mass and heat transfer, and liquid-liquid emulsification (Thakur et al., 2003; Ghanem et al., 2014; Valdés et al., 2022). On account of the multiphase flow complexity, it is very important to fully understand the characteristics of multiphase flow and the interaction between phases for designing a novel static mixer (Zidouni et al., 2015; Haddadi et al., 2020a; Haddadi et al., 2020b). Rabha et al. (2015) experimentally studied the influences of Kenics element number and superficial liquid velocity on the bubble breakup and flow pattern transition. Zidouni et al. (2015) investigated the dispersed phase volume fraction, axial velocity of gas phase and the liquid phase velocity distribution in the Kenics static mixer (KSM) using Euler-Euler method. However, there was a huge relative error between the numerical results and the experimental value because this study neglected the bubble breakup and coalescence. Meinecke et al. (2020) measured the bubble volume with the consideration of bubble deformation in SMX4 plus static mixer through particle image velocimetry (PIV) and backlight imaging

technology. [Scala et al. \(2020\)](#) also made use of PIV and backlight shadowgraph technique to evaluate the dispersion phase characteristics and the effect of mixing element number on bubble diameter evolution. [Shi et al., \(2018\)](#) investigated the effects of bubble shape variations and the turbulence induced from bubbles on BSD. [Meng et al., \(2020b\)](#) measured the BSD in TKSM and obtained the correlation among Sauter mean diameter (d_{32}), dimensionless residence time and We number.

Nowadays, the statistical characteristics of bubbles in the TKSM were studied by experimental and numerical method ([Meng et al., 2020b](#); [Yu et al., 2023](#)). However, the investigations of evolution characteristics and mechanisms of bubble breakup and coalescence in the helical static mixer are very inadequate ([Valdés et al., 2022](#)). This paper aims to investigate the bubble breakup and coalescence mechanism in TKSM through experimental and numerical methods. First, the influence of various drag forces on radial gas holdup (α_G) are evaluated and the drag correction coefficient is obtained on the basis of the experimental data. Then, the bubbles collision mechanism and the influence of coalescence correctors on calculation results are studied. Furthermore, the influence of model constant on the BSD is investigated. Finally, the effects of Reynolds number (Re) and dispersed phase volume fraction (α_d) on residence time of gas-liquid two phases are studied.

2. Experiment

The schematic diagram of the experimental system apparatus for gas-liquid mixing in TKSM is shown in [Fig. 1](#). The air compressor and multistage centrifugal pump are utilized for the delivery of gas and liquid phases, respectively. The air and water flow directions are vertically upward from the bottom of the static mixer then they flow into the circulation system. The structure and arrangement of TKSM have been described in detail in the work done by [Meng et al., \(2020b\)](#). The glass rotor flowmeter group and metal tube rotameter are separately applied to measure the volumetric flow rates of air and water. The separator is adopted to separate air and water phases for further circulation. The evolution of bubble clusters in the TKSM can be captured by the high-speed camera which has the precision of 8 bit. The capture frame rate is set as 180 FPS for high quality images and the time-scale of sampling maintains in 20 s. Data analysis methods for obtaining bubble size could take our previous publications ([Yu et al., 2022](#)) as the reference.

3. Modeling and simulation

3.1. Governing equations

The Euler-Euler model is applied to calculate the multiphase flow. [Table 1](#) shows the governing equations and closure models which includes Euler model and Turbulence model. Drag force is the foremost force among the interfacial forces ([Tas-Koechler et al., 2021](#)). Thus, the drag force must be considered in helical static mixer ([Wu et al., 2022b](#)), and the Tomiyama model was selected to evaluate the single bubble drag force ([Yu et al., 2023](#)). Taking into account of bubble swarm effect on the interfacial force, the experiment data of [Rabha et al. \(2015\)](#) was fitted to obtain a suitable drag correction coefficient for the helical static mixer. In addition to drag force, the turbulent dissipation force is also one of the most important interfacial forces in the helical static mixer ([Wu et al., 2022b](#)). The modeling of turbulent dispersion force proposed by [Burns et al. \(2004\)](#) was selected in current work. Lift force is another interfacial force that determines the radial movement of bubbles ([Zhang et al., 2020b](#)), which leads small bubbles with a diameter less than 5.8 mm to move away from the center. The presence of wall lubrication forces hinders the movement of bubbles towards the wall. Finally, the virtual mass force has the least impact on the simulation results among the interfacial force and the coefficient for virtual mass is set as 0.5 in this paper.

3.2. Turbulence modeling

Turbulence model plays a significant role in the simulation of two-phase flow. [Meng et al., \(2020a\)](#) compared the friction factor with different turbulence models in Lightnin static mixer, and they found that shear stress transport (SST) $k-\omega$ model had the highest accuracy. [Chen et al. \(2022\)](#) held the view that SST $k-\omega$ model was appropriate for predicting swirling flow. [Yu et al. \(2023\)](#) successfully utilized the SST $k-\omega$ model to calculate turbulence parameter in TKSM. Therefore, this model is applied in current study and expression is shown in Eq. (9). More details about this model could refer to the study of [Menter \(1994\)](#).

3.3. Modeling of population balance

With the intention of closing the interfacial force model and accurately describing the BSD in the static mixer, the PBM is commonly

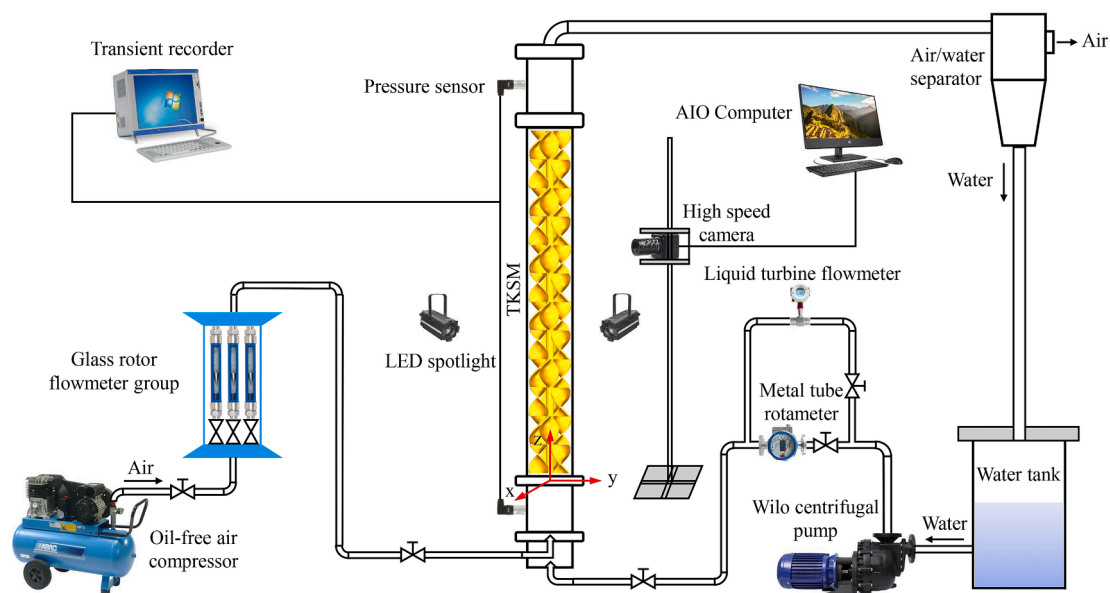


Fig. 1. Experimental setup for two-phase flow in TKSM.

Table 1

Closure models of the Euler-Euler model.

Models	Equations	
Mass conservation	$\frac{\partial}{\partial t}(\alpha_i \rho_i) + \nabla \cdot (\alpha_i \rho_i \mathbf{U}_i) = 0$	(1)
Momentum conservation	$\frac{\partial}{\partial t}(\alpha_i \rho_i \mathbf{U}_i) + \nabla \cdot (\alpha_i \rho_i (\mathbf{U}_i \mathbf{U}_i)) = -\alpha_i \nabla p + \nabla \cdot (\alpha_i \mu_{\text{eff},i} (\nabla \mathbf{U}_i + (\nabla \mathbf{U}_i)^T)) + \alpha_i \rho_i \mathbf{g} + \mathbf{F}$	(2)
	$\mathbf{F} = \mathbf{F}_D + \mathbf{F}_T + \mathbf{F}_L + \mathbf{F}_W + \mathbf{F}_{VM}$	(3)
	$\mathbf{F}_D = -\frac{3}{4} \frac{C_D}{d_b} \alpha_G \rho_L \mathbf{U}_G - \mathbf{U}_L (\mathbf{U}_G - \mathbf{U}_L)$	(4)
	$C_{D,0} = \max \left[\min \left(\frac{16}{Re_B} (1 + 0.15 Re_B^{0.687}), \frac{48}{Re_B} \right), \frac{8 Eo}{3(Eo + 4)} \right]$	
	$\frac{C_D}{C_{D,0}} = (1 - \alpha_G) f(U_L, U_G, f(U_L, U_G)) = 0.301 \left(\frac{U_L}{U_G} \right)^{0.572}$	
	$\mathbf{F}_T = -\frac{3}{4} C_D \frac{\alpha_G}{d_b} \mathbf{U}_G - \mathbf{U}_L \frac{\mu_L}{\sigma_{TD}} \left(\frac{\nabla \alpha_G}{\alpha_G} - \frac{\nabla \alpha_L}{\alpha_L} \right)$	(5)
	$\mathbf{F}_L = -C_L \rho_L \alpha_G (\mathbf{U}_G - \mathbf{U}_L) \times \text{rot}(\mathbf{U}_L)$	(6)
	$C_L = \begin{cases} \min[0.288 \tanh(0.121 Re), f(Eo_\perp)] & Eo_\perp < 4 \\ f(Eo_\perp) & \text{for } 4 < Eo_\perp < 10 \\ -0.27 & 10 < Eo_\perp \end{cases}$	
	$f(Eo_\perp) = 0.00105 Eo_\perp^3 - 0.0159 Eo_\perp^2 - 0.0204 Eo_\perp + 0.474$	
	$\mathbf{F}_W = \frac{2}{d_b} C_W \rho_L \alpha_G \mathbf{U}_G - \mathbf{U}_L ^2 \hat{\mathbf{y}}$	(7)
	$C_W(y) = f(Eo) \left(\frac{d_B}{2y} \right)^2$	
	$\mathbf{F}_{VM} = -C_{VM} \rho_L \alpha_G \left(\frac{D_G \mathbf{U}_G}{Dt} - \frac{D_L \mathbf{U}_L}{Dt} \right)$	(8)
	$C_{VM} = 0.5$	
Turbulence model	$\frac{\partial}{\partial t}(\alpha_L \rho_L k_L) + \nabla \cdot (\alpha_L \rho_L k_L \mathbf{U}_L) = \nabla \cdot \left[\left(\mu_L + \frac{\mu_t}{\sigma_{k,L}} \right) \nabla k_L \right] + \alpha_L (G_{k,L} + G_{kb,L} + S_{k,L} - Y_{k,L}) \frac{\partial}{\partial t}(\alpha_L \rho_L \omega_L) + \nabla \cdot (\alpha_L \rho_L \omega_L \mathbf{U}_L) = \nabla \cdot \left[\left(\mu_L + \frac{\mu_t}{\sigma_{\omega,L}} \right) \nabla \omega_L \right] +$	(9)
	$\alpha_L (G_{\omega,L} + G_{\omega b,L} + S_{\omega,L} - Y_{\omega,L}) \mu_t = \frac{\rho k_L}{\omega_L} \frac{1}{\max \left[\frac{1}{\alpha^*}, \frac{SF_1}{\alpha_1 \omega_L} \right]} F_1 = \tanh(\phi_1^4) \phi_1 = \max \left(\frac{2\sqrt{k_L}}{0.09 \omega_L y}, \frac{500 \mu_L}{\rho y^2 \omega_L} \right)$	

applied to describe the breakup and coalescence phenomena of bubbles or droplets.

The transport equation of the number density functions for bubble size is shown in Eq. (10).

$$\frac{\partial}{\partial t}[n(v)] + \nabla \cdot [U_G n(v)] = B_{Br} - D_{Br} + B_{Ag} - D_{Ag} \quad (10)$$

where B_{Br} , D_{Br} , B_{Ag} and D_{Ag} denote the terms of bubble birth and death

the bubble would break if the kinetic energy contained in turbulent eddy exceeded the surface energy increment of the bubble. [Lehr et al. \(2002\)](#) supposed that the bubble would break if inertial force of the turbulent eddy was more than the interface force of the tiniest particle. Similar to the structure of the Luo breakup model, they also gave the product form of the collision frequency and breakup probability of bubbles. The specific expression of the Luo breakup model and Lehr breakup model could be obtained from Eq. (11).

$$\begin{aligned}
 & \text{Common part of two models} \\
 & b(f_v, d) = K \int_{\xi_{\min}}^1 \frac{(1+\xi)^2}{\xi^n} \exp(-b\xi^m) d\xi, \xi = \frac{\lambda}{d} \\
 & \beta(f_v, d) = \frac{2b(f_v, d)}{\int_0^1 b(f_v, d) df_v} \\
 & \text{Luo breakup model} \quad K = 0.9238 \epsilon^{1/3} d^{-2/3} \alpha_L, b = 12 c_f \sigma \rho_L^{-1} \epsilon_L^{-2/3} d^{-5/3} \beta^{-1} \\
 & \quad c_f = f^{2/3} + (1-f)^{2/3} - 1, f = V/V', m = -11/3, n = 11/3, \beta = 2.047 \\
 & \text{Lehr breakup model} \quad K = 1.19 \epsilon^{-1/3} d^{-7/3} \sigma \rho_L^{-1} f^{-1/3} \\
 & \quad b = 2 We_{crit} \sigma \rho_L^{-1} \epsilon^{-2/3} d^{-5/3} f^{-1/3}, m = -2/3, n = 13/3, We_{crit} = 0.1
 \end{aligned} \quad (11)$$

caused by bubble breakup and coalescence, respectively.

[Luo and Svendsen \(1996\)](#) proposed the breakup model under the assumption that the single bubble always occurred binary breakup and the turbulence remained isotropic. Meanwhile, they held the view that

where $\xi_{\min}$ equals to 11.4 times as much as Kolmogorov scale, ϵ represents turbulent dissipation rate, We_{crit} is the critical Weber number, ξ equals to λ/d and σ represents surface tension.

[Table 2](#) summaries the equations for bubble coalescence kernels used

Table 2
Equations of bubble coalescence.

Item	Equations
Bubble coalescence rates	$a(v_i, v_j) = M_c C_{ij} P_c(d_i, d_j)$ (12)
Bubbles Collision frequency	Due to turbulent fluctuation $C_{ij}^T = \frac{\pi}{4} (d_i + d_j) \sqrt{2\varepsilon}^{1/3} (d_i^{2/3} + d_j^{2/3})^{1/2}$ (13)
	Due to buoyancy driven $C_{ij}^B = \frac{\pi}{8} (d_i + d_j)^2 C_{buoy} u_{bi} - u_{bj} $ (14)
Due to wake entrainment	$u_{bi} = \left(\frac{2.14\sigma}{\rho_L d_i} + 0.505gd_i \right)^{0.5}, C_{buoy} = 1$
	$C_{ij}^W = \frac{\pi}{8} (d_i^2 \theta_i u_{wi} + d_j^2 \theta_j u_{wj}), u_{wi} = u_{bi} C_D^{1/3}$ (15)
Due to viscous shear	$\theta_i = \begin{cases} \frac{(d_i - 0.5d_{crit})^6}{(d_i - 0.5d_{crit})^6 + (0.5d_{crit})^6} & d_i \geq 0.5d_{crit} \\ 0 & \text{else} \end{cases}$
	$d_{crit} = 4.0 \sqrt{\frac{\sigma}{g(\rho_L - \rho_G)}}$ $C_{ij}^S = 0.5 \frac{\pi}{4} (d_i + d_j)^2 \left[\frac{0.5}{\pi} (d_i + d_j) \gamma_b \right] \gamma_b =$ (16)
Prince & Blanch coalescence model	$\left(2 \left(\frac{\partial u_L}{\partial x} \right)^2 + 2 \left(\frac{\partial v_L}{\partial y} \right)^2 + 2 \left(\frac{\partial w_L}{\partial z} \right)^2 + \left(\frac{\partial u_L}{\partial y} + \frac{\partial v_L}{\partial x} \right)^2 + \left(\frac{\partial u_L}{\partial z} + \frac{\partial w_L}{\partial x} \right)^2 + \left(\frac{\partial v_L}{\partial z} + \frac{\partial w_L}{\partial y} \right)^2 \right)^{1/2}$
	$P_c(d_i, d_j) = \exp \left(- \frac{r_{ij}^{5/6} \rho_L^{1/2} \varepsilon^{1/3}}{4\sigma^{0.5}} \ln \frac{h_0}{h_f} \right)$ (17)
Luo coalescence model	$h_0 = 10^{-4} \text{m}, h_f = 10^{-8} \text{m}$ $r_{ij} = \frac{d_i d_j}{d_i + d_j}$
	$P_c(d_i, d_j) = \exp \left\{ - \frac{[0.75(1 + d_i^2/d_j^2)(1 + d_i^3/d_j^3)]^{1/2}}{(\rho_G/\rho_L + 0.5)^{1/2}(1 + d_i/d_j)^3} We_{ij}^{1/2} \right\}$ (18)
Liao et al. coalescence model	$u_{rel} = 1.43\varepsilon^{1/3} (d_i^{2/3} + d_j^{2/3})^{1/2}, We_{ij} = \frac{\rho_L d_i (u_{rel})^2}{\sigma}$
	$P_c(d_i, d_j) = \exp \left(- 2.5 We_{ij}^{0.5} \right)$ (19)
Turbulence model	$We_{ij} = \frac{\rho_L d_{eq}}{\sigma} u_{rel}$ $d_{eq} = \frac{2d_i d_j}{d_i + d_j}$
	$a(v_i, v_j) = 1.294 \zeta_r \gamma \frac{(d_i + d_j)^3}{8} \gamma = \left(\frac{\varepsilon}{v_L} \right)^{0.5}$ (20)
Turbulence model	Collision rate (Bubble coalescence in the viscous subrange)
	Collision rate (Bubble coalescence in the inertial subrange) $a(v_i, v_j) = 5.012 \zeta_r \frac{(d_i + d_j)^2}{4} \sqrt{(u_i^2 + u_j^2)} \zeta_r = 0.732 \left(\frac{5}{N_T} \right)^{0.242} N_T = \frac{6\pi\mu(d_i + d_j)^3 \lambda_q \lambda_q}{8H}$ (21)
Das model	$\left(\frac{4\varepsilon}{15\pi v_L} \right)^{0.5}$
	$a(v_i, v_j) = \frac{\sqrt{8\pi R_{ij}^2}}{\varphi} \sqrt{\sigma_{U_{d_i}}^2 + \sigma_{U_{d_j}}^2} \left[1 - \exp \left(\frac{-0.5U_{cr}^2}{\sigma_{U_{d_i}}^2 + \sigma_{U_{d_j}}^2} \right) \right]$ (22)
Yang and Xiao	$\sigma_{U_{d_i}} = C_{Das} \frac{\pi^{0.5} (\varepsilon d_i)^{1/3}}{4}, U_{cr} = 2\pi \sqrt{\frac{0.5\sigma}{3\rho_L d_{eq}}}, R_{ij} = 0.5(d_i + d_j)$
	$M_c = \begin{cases} 0.29733 + 1.25303 \exp \frac{U_G}{0.0219}, & U_G \leq 0.101 \text{m/s} \\ 0.21454 + 1.04371 \exp \frac{U_G}{0.06278}, & U_G > 0.101 \text{m/s} \end{cases}$ (23)
Wang and Wang	$M_c = \begin{cases} \frac{\alpha_{max}}{\alpha_{max} - \alpha_G}, & \alpha_{max} = 0.8 \end{cases}$ (24)

in this paper. The bubble coalescence model is composed of collision frequency and coalescence efficiency. Collision frequency could mainly be induced by the effects of turbulent fluctuation, viscous shear, wake entrainment and buoyancy driven (Liao and Lucas, 2010; Zhang and Luo, 2020). At present, turbulent fluctuation has been widely deemed as the main factor inducing bubble coalescence. However, the influences of wake entrainment, buoyancy driven and viscous shear mechanisms on bubble coalescence behavior are rarely studied. Zhang and Luo (2020) found that the viscous shear mechanism overestimated bubble coalescence rate in bubble column. The calculation methods of coalescence efficiency can mainly be divided into energy models and film drainage models. Among them, the film drainage mechanism is currently the

mainstream viewpoint to judge bubble coalescence efficiency. There are two characteristic time scales: contact and drainage time, and they are employed by this model to determine the bubble coalescence efficiency. The aforesaid viewpoint was also adopted by researchers to establish corresponding coalescence models (Prince and Blanch, 1990; Luo, 1993). In addition, the turbulence model does not provide specific expressions for coalescence efficiency and collision probability. The turbulence model gives different coalescence expressions in the viscous and inertial subranges, and the coalescence rate of bubbles can be directly calculated (Abrahamson, 1975; Saffman and Turner, 1956). Similar to the structure of turbulence model, the Das model directly considered the turbulence influence on bubble coalescence rate and the impact of

contact angles on bubble collisions (Das, 2015). In addition, it can be widely applied to different flow conditions due to the existence of an empirical constant C_{Das} in this model (Zhang and Luo, 2020).

The closures for the breakup and coalescence phenomena of adiabatic bubbly flows had been defined as baseline model (Liao et al., 2015). The coalescence and breakup models considering entire turbulent spectrum could describe the breakup and coalescence rate of bubbles more accurately (Han et al., 2014; Zhang et al., 2023a). However, it had been proven that the breakage of bubbles was independent on the turbulent spectrum when the liquid phase viscosity was below 8 mPa•s (Zhang et al., 2023b). Furthermore, advanced models that consider entire turbulent spectrum have higher computational costs and lower computational stability. In addition, the experiment found that the main reason for the breakage of bubbles in the swirl flow field was the collision between the turbulence eddies induced by the elements and the bubbles (Wu et al., 2022; Maluta et al., 2023). Therefore, the direct bubble wall interactions are not considered in this paper. We attempt to use different coalescence and breakup models which are applicable within the inertial subrange to study dynamic behavior of bubbles in TKSM.

3.4. Settings for simulation process

3.4.1. Boundary conditions

The turbulence characteristics and bubble dynamics behavior are solved using the CFD software ANSYS Fluent. The boundary conditions are shown in Table 3. The inlets of gas and liquid phase are selected as the velocity boundary condition. Hydraulic diameter (D_H) and turbulence intensity are employed to describe the turbulence conditions at inlet position. The boundary condition for outlet is set as outflow and no slip boundary condition is employed for all wall surface. The maximum and minimum values for bubble diameters are set as 12.50 mm and 0.16 mm, respectively. The diameter is determined by the Eq. (25) and the q in this paper is equal to 1 (Lister et al., 1995). Azizi and Al Taweel (2011) found that the initial dispersed phase size had a little effect on the final droplets size after seventh mixing elements. The d_{32} is applied to characterize the distribution of bubbles and the specific expression is shown in Eq. (26) (Xu et al., 2023). The final d_{32} comes to 3.72 mm and 3.99 mm, respectively, when the initial bubble diameter is 9.92 mm and 12.50 mm. The relative deviation between the final d_{32} is 6.76 %. That means the initial bubble size is not a significant parameter for calculation results after the disturbance of mixing elements. Therefore, 9.92 mm is used as the initial bubble size in this paper. The momentum equations as well as the other equations are discretized through first-order upwind scheme (Das, 2015; Wei et al., 2018) and QUICK scheme, respectively. The phase-coupled SIMPLE approach is applied for the solution of pressure-velocity coupling. The time step size in the range of 1.742×10^{-4} to 5.805×10^{-4} s is derived from CFL condition, i.e., $0.3 < \text{Courant} < 1$ (Wei et al., 2018). As a result, the time step size for instantaneous calculation is 5×10^{-4} s. The residual criterion for activated equations has been selected as 10^{-4} .

Table 3
Boundary conditions.

Parameters	Boundary conditions
Inlet velocity of liquid phase	$u_c = \frac{(2m_0 + 1)(m_0 + 1)}{2m_0^2} \left(1 - \frac{\sqrt{x^2 + y^2}}{D_0/2} \right)^{1/m_0} \bar{u}_c$
Inlet velocity of gas phase	$u_d = \frac{\bar{u}_d(w + 2)}{w} \left(1 - \left(\frac{\sqrt{x^2 + y^2}}{d_0/2} \right)^w \right)$
Hydraulic diameter	$D_H = \frac{4A}{P}$
Turbulence intensity	$I = 0.16Re^{-1/8}$
Outlet	Outflow
Wall	No slip

Table 4

Grid independence test using turbulent dissipation rate.

Position	$z/l = 1$	$z/l = 13$
N_1, N_2, N_3	2365640, 2122167, 1724841	
r_{21}	1.2245	
r_{32}	1.2205	
p	20.2076	16.3617
$e_a^{21}(\%)$	0.32	0.39
$e_{ext}^{21}(\%)$	0.32	0.40
$GCI_{fine}^{21}(\%)$	0.68	1.83

$$\frac{V_{i+1}}{V_i} = 2^{1/q} \quad (25)$$

$$d_{32} = \frac{\sum (n_i d_i^3)}{\sum (n_i d_i^2)} \quad (26)$$

3.4.2. Mesh and bin number independence.

Compared with other kinds of grids, the use of polyhedral meshes can improve the computational efficiency and ensure the accuracy of calculation results. The turbulent dissipation rate (ϵ) is chosen as a second order turbulence quantity for analyzing mesh independence. Table 4 describes the calculated GCI using ϵ for different types of grids at $z/l = 1$ and $z/l = 13$. Calculation results indicate that the uncertainty of these three types of grids is all below 2 %. The refined mesh with 10 boundary layers was sufficient to ensure the calculation accuracy (Meng et al., 2020a; Meng et al., 2021).

As is known to all, turbulence quantities directly determine the accuracy of PBE solution and the bubble or droplet breakup rate is affected by turbulence intensity (Zhou et al., 2022). Fig. 2(a) shows the radial distribution of turbulence intensity for different types of grids. The turbulence intensity has a peak value at the center, which is caused by the fully developed turbulence at the center of the smooth tube. Then turbulence intensity decreases with the rise of the radial distance, which could attribute to the decrease of velocity. It has a rapid increase and finally achieves its maximum near wall surface, which may be induced by the secondary flow of liquid phase and wake acceleration effect of bubbles. The large sized bubble at the tube wall moves toward the tube center and the small sized bubble at the center diffuses toward the wall surface. Based on the calculation results of grid numbers 2365640, the mean relative deviations (MRE) of turbulence intensity are 6.85 %, 3.85 %, 1.87 % and 1.72 %, respectively. To sum up, the mesh N_2 will be utilized in the following work.

Discrete method has been employed to solve PBEs in this work due to its better robustness. The bin number independence validation is presented in Fig. 2(b). By analyzing the calculation results, the maximum deviation of α_G between 16 bins and 25 bins is 7.27 % and the maximum deviation of α_G between 20 bins and 25 bins is only 1.59 % as shown in Fig. 2(b). It should be noted that the selection of bin number is crucial for the precision of the calculation results. The bin number is set as 20 in the following work by comprehensively considering the calculation accuracy and efficiency.

4. Results and discussion

4.1. The influence of drag models

The swirling flow and radial secondary flow induced by mixing elements have great influence on gas holdup distribution in static mixers. The three-dimensional distribution of gas holdup is shown in Fig. 3. The entire cross-section flow field is composed of many vortices. As a result, bubbles will gather around the elements due to the entrainment effects of vortices and swirling flow fields, which makes the gas holdup near the element larger.

The α_G is a very important flow parameter in gas-liquid reactors. It

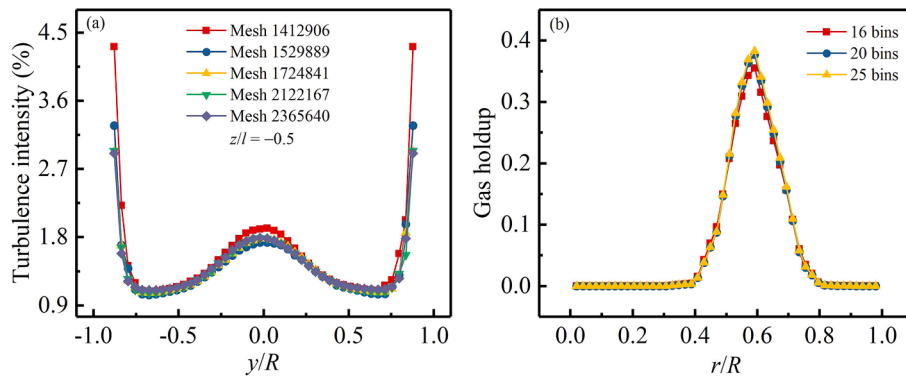


Fig 2. Grid and bin number independence tests based on (a) turbulence intensity and (b) gas holdup.

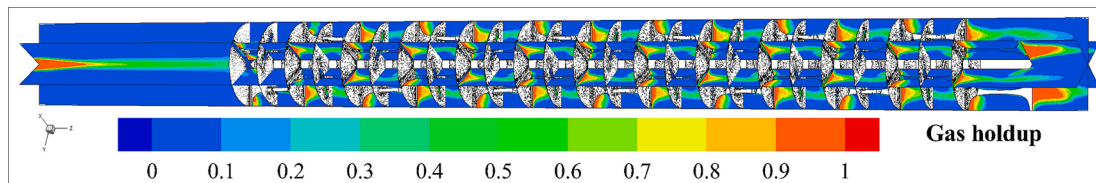


Fig 3. The radial secondary flow and gas holdup distribution at different axial positions in TKSM.

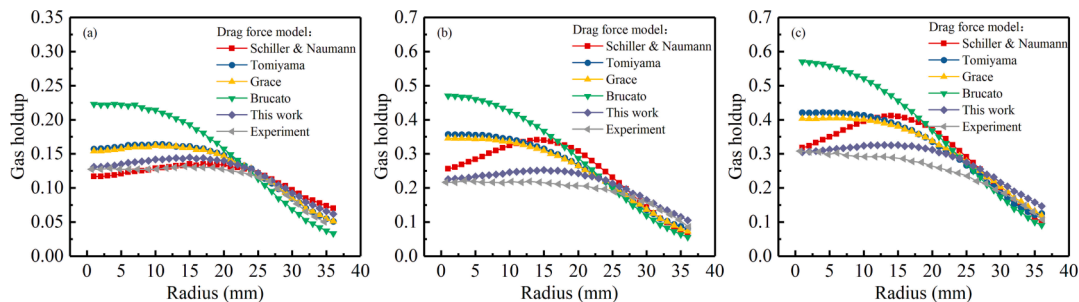


Fig 4. The influence of various drag models on radial α_G and comparison between simulation and the experiment of Rabha et al. (2015) at (a) $U_G = 0.11$ m/s, (b) $U_G = 0.23$ m/s and (c) $U_G = 0.32$ m/s.

directly determines the liquid swirling flow in the static mixer, which has a great effect on the mixing efficiency and two-phase mass transfer. Fig. 4 compares the difference between the radial α_G obtained by several drag models and the experimental value under various superficial gas velocity (U_G) when the superficial liquid velocity (U_L) equals to 0.6 m/s. It can be seen from Fig. 4 that the α_G reaches its maximum at the center and gradually descends with increasing radial distance. Fig. 4(a) shows that Schiller-Naumann model provides better agreement with experiment data than that of Tomiyama and Grace models when r/R is less than 0.625 under lower superficial gas velocity. Small bubbles are not easy to coalesce into large bubbles at lower U_G . It is because Schiller-Naumann model (Schiller and Naumann, 1933) does not consider bubble deformation ($d_b < 2.5$ mm) while Tomiyama models (Tomiyama et al., 1998) utilized Eo number and Grace models (Grace et al., 1976) adopted Eo number and Mo number to characterize bubble deformation. The excessive gas holdup predicted by Tomiyama and Grace models reveals that the bubble deformation is overestimated. The Brucato model (Brucato et al., 1998) additionally considered the influence of microscale turbulence on the movements of bubble clusters, which induces the Brucato model to severely overestimate the radial gas holdup when r/R is less than 0.625. It is worth noting that the Brucato model also promotes the liquid swirling flow in the static mixer, so that the α_G near wall surface is smaller than the experimental value. In this work, we took into account the large bubbles wake effect and used two-phase

velocity ratio as a correction factor to modify the Tomiyama model. In contrast with the α_G of experimental values, the MRE of the Schiller-Naumann, Tomiyama, Grace, Brucato and modified models are 13.15 %, 15.34 %, 14.26 %, 32.62 % and 7.33 %, respectively. It can be found that the modified model could significantly improve the prediction accuracy.

In addition, the local α_d in the pipe has an uptrend and the radial gas holdup also increases with the rise of U_G . Fig. 4(b and c) separately show the radial gas holdup distribution when U_G are 0.23 m/s and 0.32 m/s. The Brucato model always overestimates the radial α_G at the center, which is similar to the prediction result of $U_G = 0.11$ m/s. Meanwhile, bubble induced turbulence (BIT) effect and coalescence rate are enhanced with the rise of U_G , and the bubble deformation effect dominates the expression of drag force. The Grace as well as Tomiyama model have slightly improved their prediction accuracy. In comparison with the experimental values of $U_G = 0.32$ m/s, the MRE of the Schiller-Naumann, Tomiyama, Grace, Brucato and modified models are 14.76 %, 26.80 %, 26.68 %, 48.19 % and 12.23 %, respectively.

4.2. Effects of bubble breakup and coalescence kernels

Fig. 5 demonstrates the axial d_{32} curves obtained from different combinations of breakup and coalescence kernels and experimental measurements. The bubble size gradually descends with the rise of the

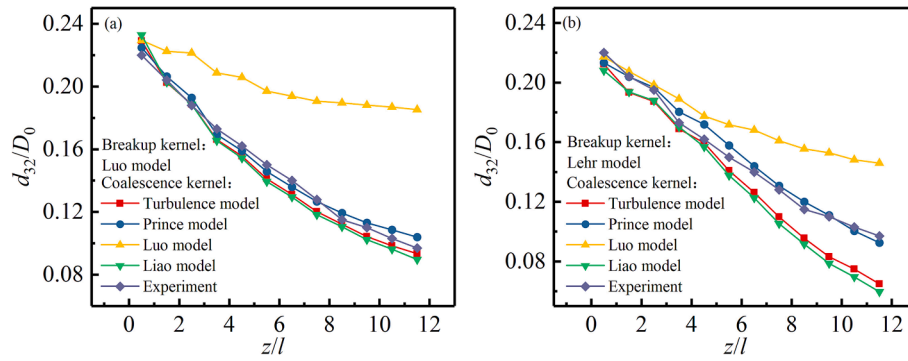


Fig 5. Comparison of axial d_{32} for different model combinations (a) when breakup kernel is Luo, (b) when breakup kernel is Lehr under $Re_L = 33000$ and $\alpha_d = 0.05$.

flow length as shown in Fig. 5. The high shear forces generated by mixing components and turbulent eddy jointly promote the bubble breakup and inhibit the generation of large bubbles. As clearly shown in Fig. 5, the turbulence coalescence model and Liao coalescence model always underestimates the axial d_{32} when the breakup kernel is the Luo model and Lehr model. On the contrary, the Luo coalescence model seriously overestimates the axial d_{32} . Prince coalescence model gives better prediction results. In comparison with the experimental results, the MRE of axial d_{32} of Luo-Prince model and the Lehr-Prince model are 2.86 % and 3.07 %, respectively. In comparison with the axial d_{32} obtained by Lehr-Prince model, the prediction accuracy of Luo-Prince model is improved by 6.84 %.

In addition, the axial d_{32} predicted by the Lehr model is slightly smaller than that of Luo model as observed in Fig. 5. This may be because the bubble breakup rate predicted by Lehr model exceeds that of Luo model. The bubble breakup rate calculated by the two breakup kernels are depicted in Fig. 6. The predicted results of both breakup kernels show that the breakup rate of bubbles will increase with the rise of its size and ε . However, the breakup rate of bubbles obtained by Luo model is obviously smaller than that of Lehr model. The breakup rate of the Lehr model almost shows an exponential growth trend, while that of the Luo model shows an obvious linear increase. The bubble breakup rate obtained by Luo model is obviously less than that of Lehr model. Therefore, d_{32} obtained by Luo breakup model is lower than the prediction result of Lehr breakup model.

The shear forces are induced by the variation of velocity gradient in the swirling flow field. In order to distinguish the shear force between inlet and mixing section, the shear force at different axial positions is depicted in Fig. 7. The shear force of the empty pipe section mainly

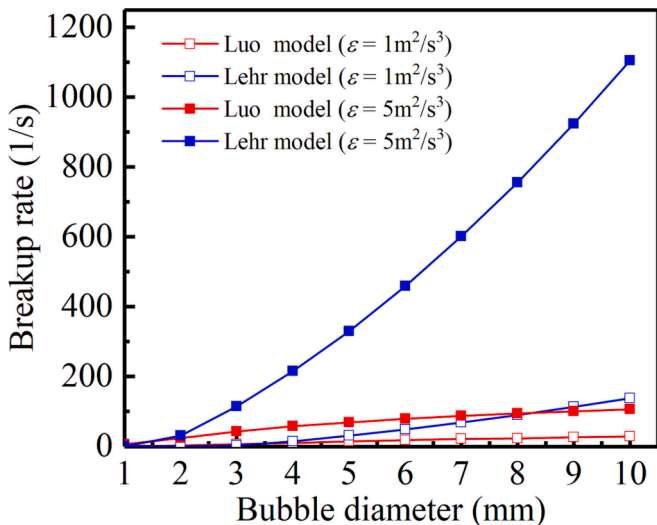


Fig 6. Breakup rate of Luo and Lehr models.

occurs at the tube wall. The velocity gradient increases due to the cutting effect of the elements and the shear force increases when the fluid flows through the leading edges of the mixing elements. With the increase of axial distance, the higher shear force occurs around the surface of elements and shear force around elements is gradually strengthened due to the velocity gradient induced by vortices. The shear force at $z/l = 1$ exceeds that at $z/l = 0$. The high shear force occurs near elements and wall, which is similar to the conclusions obtained by Hobbs et al. (1998).

Fig. 8 demonstrates the BSD and CPD of the section at $z/l = 11.5$. All tested models in this work show bimodal distributions. This can be attributed to two aspects. The first is the bubble deformation caused by high shear force which is generated from TKSM elements. The second is the internal flow mechanism of bubbles produced by the density discrepancy of air and water.

Meanwhile, these two peaks represent that the small bubbles hindering effect and the large bubbles accelerating effect jointly control the flow characteristics in TKSM. It can be clearly concluded from Fig. 8(a) that the Luo-Turbulence and Luo-Liao models underestimate the diameter corresponding to the peak of BSD. However, Luo-Luo models overestimate the diameter corresponding to the peak of BSD. The calculation results of the Luo-Prince model are the most appropriate to the experimental results. In this section, the Luo coalescence model, Prince coalescence model and Liao coalescence model adopt the same coalescence mechanism which is called turbulent fluctuation. Therefore, the difference of simulation results is caused by the various coalescence efficiency equations. In contrast with experimental data, the MRE of CPD curves shown in Fig. 8(b) obtained by Luo-Turbulence, Luo-Prince, Luo-Luo and Luo-Liao models are 23.57 %, 20.93 %, 40.60 % and 27.89 %, respectively. The Luo-Prince model has the best prediction results.

Fig. 9 describes the coalescence efficiency profiles for different coalescence models. When bubble diameter is below 1 mm, the coalescence efficiency obtained by Prince model overtakes that of Luo model as can be seen in Fig. 9. The coalescence efficiency of Luo coalescence model exceeds that of Prince model with the increase of bubble collision diameter. As a result, the peak value of the BSD curves of small bubbles calculated by Luo coalescence model is significantly larger than the peak value calculated by Prince model. Because the Luo model underestimates the coalescence efficiency of small bubbles, small bubbles can easily gather into large bubbles rather than disperse into smaller bubbles. It reveals that the number of minute bubbles calculated by Luo coalescence model exceeds that of Turbulence, Prince and Liao model. The calculation result of Liao model has the minimum coalescence efficiency, so the predicted bubble size of Liao model is the smallest. The calculated d_{32} of Turbulence coalescence model is the second smallest.

Fig. 10 describes the bubble coalescence rate calculated by turbulence model. The bubble movement speed and bubble coalescence rate increase in TKSM with the increase of ε . In addition, the coalescence rate has an uptrend as the diameter of collision bubbles increases. At present, turbulence coalescence model had been successfully applied in multi-phase flow of bubbles and hydrate particles (Cao et al., 2022). However,

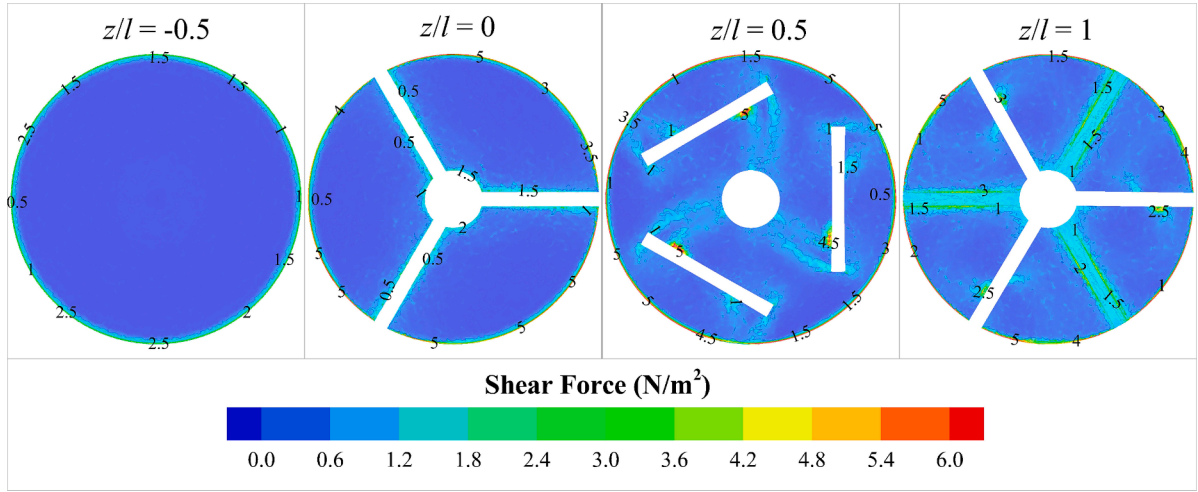


Fig 7. The contour of shear force at different axial positions.

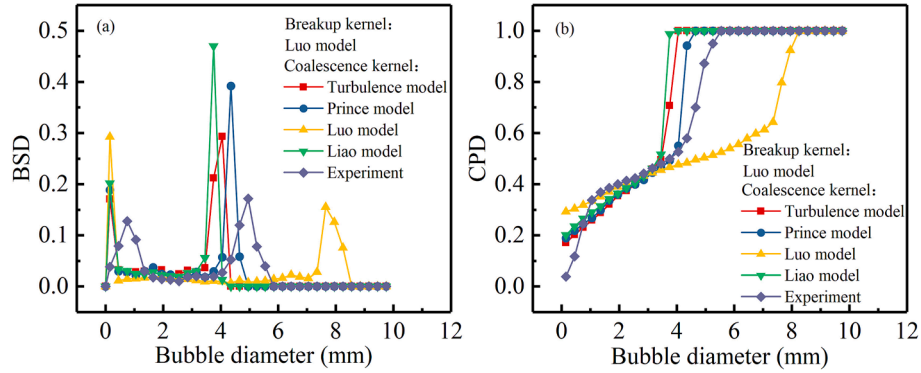


Fig 8. Effects of various combinations of coalescence kernel and Luo breakup kernel on (a) BSD and (b) CPD under $Re_L = 33000$ and $\alpha_d = 0.05$.

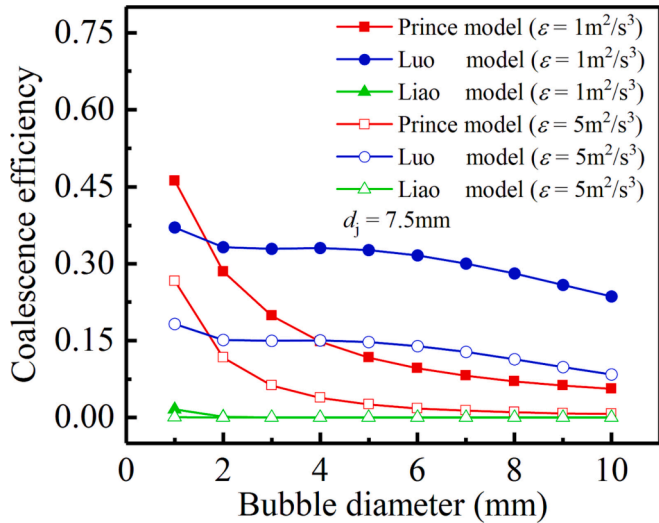


Fig 9. Coalescence efficiency calculated by Prince, Luo and Liao models.

almost no bubble coalescence occurs in viscous subrange. Compared the Fig. 10 with Fig. 6, the coalescence rate calculated by turbulence model is so low that it cannot match well with the bubble breakup rate predicted by Luo model and Lehr model. It could be understood that only breakup is possible for large sized bubbles in a shear dominated flow regime.

Fig. 11 depicts the BSD and CPD predicted by Lehr-Turbulence, Lehr-Prince, Lehr-Luo and Lehr-Liao models. It can be observed from Fig. 8(a) and Fig. 11(a) that the BSD curve predicted by Lehr breakup model moves to the left. That also means that the breakup capacity of Lehr model overtakes that of Luo model. Overall, the trend in BSD curves predicted by two breakup models is basically the same. However, the number of small bubbles predicted by Luo breakup model is higher than that of Lehr model. This is mainly due to the difference between their dimensionless size distribution function.

Fig. 12 describes the daughter size distribution curves of Lehr and Luo breakup model, respectively. The daughter size distribution curve calculated by Lehr model is Λ shape, and the curve obtained by Luo model presents a U-shaped distribution. Therefore, Lehr model tends to occur equal size breakage and Luo model tends to have unequal size breakage. Andersson and Andersson (2006) found that droplets were prone to occur equal size breakage, while bubbles were more likely to occur unequal size breakage. As depicted in Fig. 11(b), the MRE between the predicted CPD of Lehr-Turbulence, Lehr-Prince, Lehr-Luo and Lehr-Liao models and the experimental value is 35.17 %, 22.77 %, 32.46 % and 38.78 %, respectively. The results clearly reveal that Lehr-Prince model could give a more desirable simulation result.

Overall, Luo model always overestimates the bubbles coalescence rate in TKSM. The breakup models of both Luo and Lehr predict the bimodal distribution and underestimate the quantity of small bubbles in TKSM. The Luo-Prince model gives the most accurate results. Therefore, it is adopted as the basic model in this paper. The BSD, ϵ and axial d_{32} curves predicted by Luo-Prince model are shown in Fig. 13. It can be clearly seen in Fig. 13(a) that the peak value of BSD predicted by Luo-

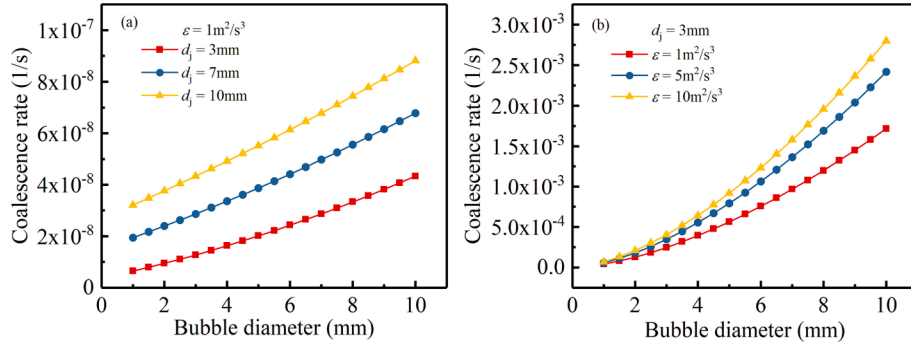


Fig 10. Bubble coalescence rate calculated by turbulence model (a) influence of bubble collision diameter and (b) influence of ε .

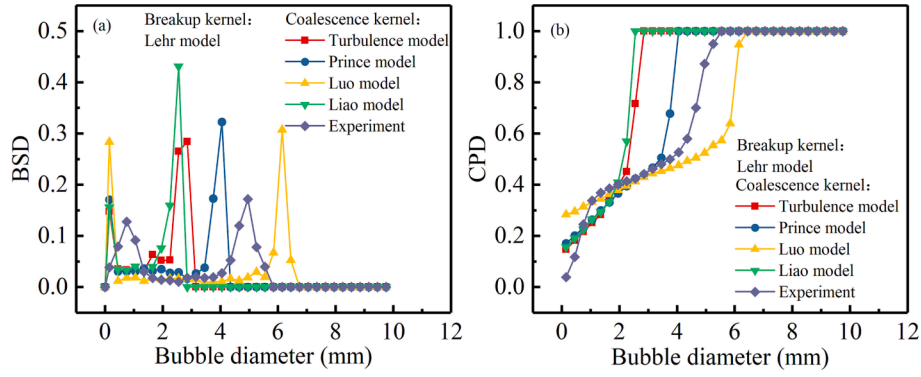


Fig 11. Effects of various combinations of coalescence kernel and Lehr breakup kernel on (a) BSD and (b) CPD under $Re_L = 33000$ and $\alpha_d = 0.05$.

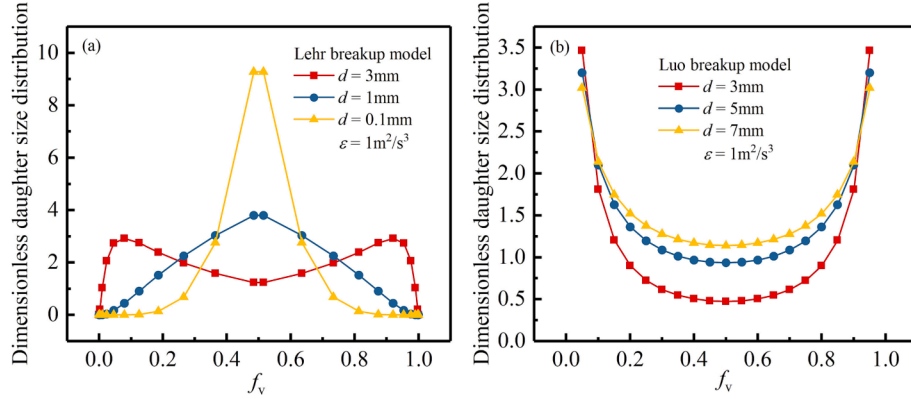


Fig 12. Daughter size distribution calculated by Lehr and Luo breakup models.

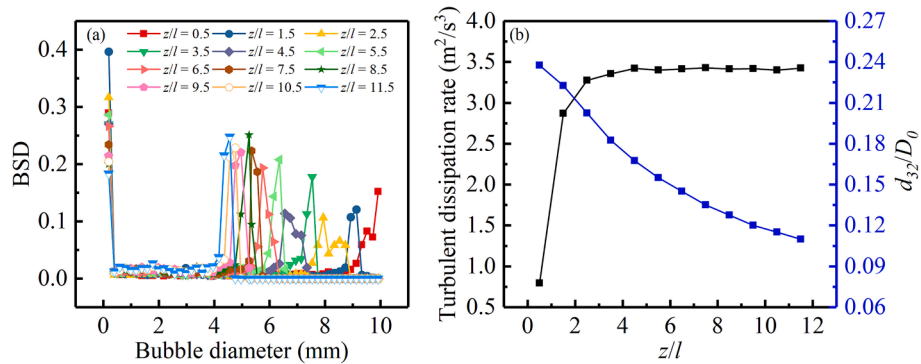


Fig 13. The (a) BSD and (b) turbulent dissipation rate and axial d_{32} at different axial positions.

Table 5
Meanings of Case 1 – Case 4.

Model	Investigated bubbles collision mechanisms
Case 1	$C_{ij} = C_{ij}^T$
Case 2	$C_{ij} = C_{ij}^T + C_{ij}^B$
Case 3	$C_{ij} = C_{ij}^T + C_{ij}^B + C_{ij}^W$
Case 4	$C_{ij} = C_{ij}^T + C_{ij}^B + C_{ij}^W + C_{ij}^S$

Prince model shifts to left with the increase of element numbers. The large sized bubbles continue to break into smaller bubbles and axial d_{32} gradually decreases during the flow process. This maybe because the ε first increases then tends to be stable with the increasement of axial positions after four elements as shown in Fig. 13(b).

4.3. Effects of bubble coalescence mechanisms and coalescence correction coefficient

The bubbles collision mechanisms are considered as depicted in Table 5. Fig. 14 shows the BSD and CPD predicted by different bubble coalescence mechanisms. According to Fig. 14(a), the diameter corresponding to the peak of the predicted BSD in Case 4 is slightly greater than experimental value and the BSD prediction results of Case 1, 2 and 3 are almost consistent. The MRE of d_{32} predicted by four cases are 3.45 %, 3.41 %, 3.47 % and 12.27 %, respectively. It indicates that the most foremost collision mechanism of bubbles is turbulent fluctuation and the bubble collision frequency caused by viscous shear is overestimated. Azizi et al. (2022) found that the mixing element could efficiently improve the velocity gradient and energy dissipation rate in SMX-Plus static mixers. Therefore, the reason for the increase of bubble size may be that the mixing element augments the bubble collision frequency. However, the coalescence rate is too small to match the breakup rate.

Fig. 15 shows the effects of turbulent fluctuation, viscous shear, wake entrainment and buoyancy driven on bubble collision frequency. Viscous shear becomes the second most important factor affecting bubble collision. The wake entrainment and buoyancy driven effects almost have no effect on the bubble collision when the bubble diameter is less than 9 mm as also shown in Fig. 14. The collision frequency redoubles as the shear rate increases from 100 (1/s) to 200 (1/s). In contrast to experimental values, the MRE of CPD calculated by Case 1, 2, 3, and 4 are 20.93 %, 21.34 %, 20.74 % and 23.01 %, respectively. In summary, the prediction result of Case 3 is acceptable. Turbulent fluctuation is the main factor leading to the coalescence behavior of bubbles, following by buoyancy driven and wake entrainment. The viscous shear mechanism overestimates the bubble collision frequency in TKSM. Meanwhile, the similar conclusions were found in the bubble column (Zhang and Luo, 2020).

Fig. 16 demonstrates the influence of coalescence correctors to bubble diameter by considering three and four coalescence mechanisms,

respectively. There is an obvious phenomenon in Fig. 16(a) that the coalescence rate increases and the BSD peak moves to the right when Wang coalescence coefficient is used. This can be attributed to the expression of Wang and Wang (2007) coalescence correctors. The expression in Table 2 indicates that the coalescence rate will augment as the local α_G rises and reach the maximum when $\alpha_G = 0.8$. Compared with experimental values, the MRE of CPD calculated by Prince3-Wang and Prince4-Wang models are 21.22 % and 25.04 %, respectively. The coalescence correctors of Yang and Xiao derived from the EMMS method (Yang and Xiao, 2017). Yang and Xiao held the view that the coalescence correctors were only relevant to the U_G and the coalescence rate descended with the rise of the U_G when U_G was less than or equal to 0.101 m/s. The corresponding coalescence correction coefficient is 0.3822 at $Re_L = 33000$ and $\alpha_d = 0.05$, which reduces the coalescence rate of bubbles in TKSM. Fig. 16(a) clearly demonstrates that the use of Yang and Xiao coalescence correctors reduces the bubble size and makes the BSD peak shift to the left. Compared with experimental values, the MRE of CPD calculated by Prince3-Yang and Prince4-Yang models are 23.17 % and 19.04 %, respectively. The contribution of viscous shear to bubble coalescence is proved again due to the existence of mixing elements. The contribution of buoyancy driven and wake entrainment to bubble coalescence can be almost ignored in TKSM. The coalescence kernel considering turbulent fluctuation, viscous shear, buoyancy driven, wake entrainment and EMMS model could be used to describe the BSD in TKSM more accurately.

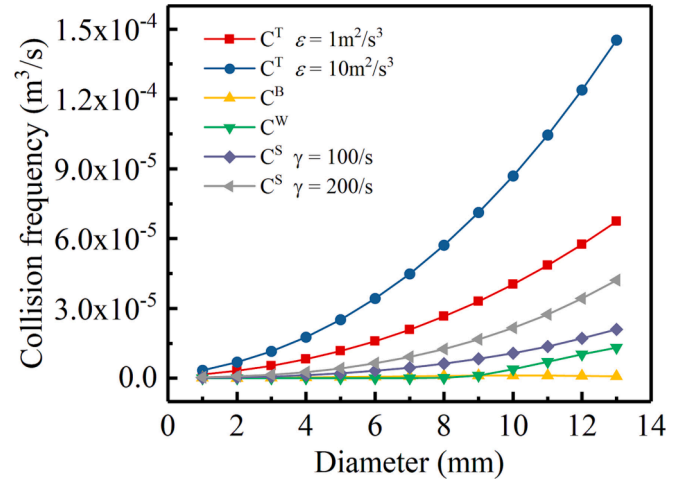


Fig 15. The effect of different bubble coalescence mechanisms on collision frequency.

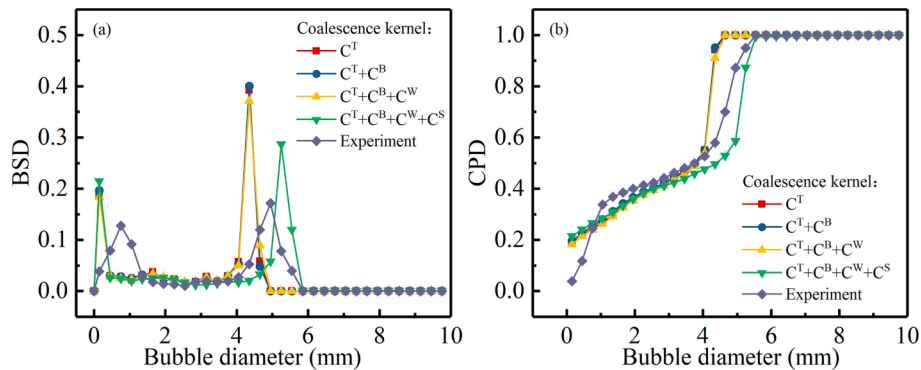


Fig 14. Effects of different bubble collision mechanisms on (a) BSD and (b) CPD at $z/l = 11.5$ under $Re_L = 33000$ and $\alpha_d = 0.05$.

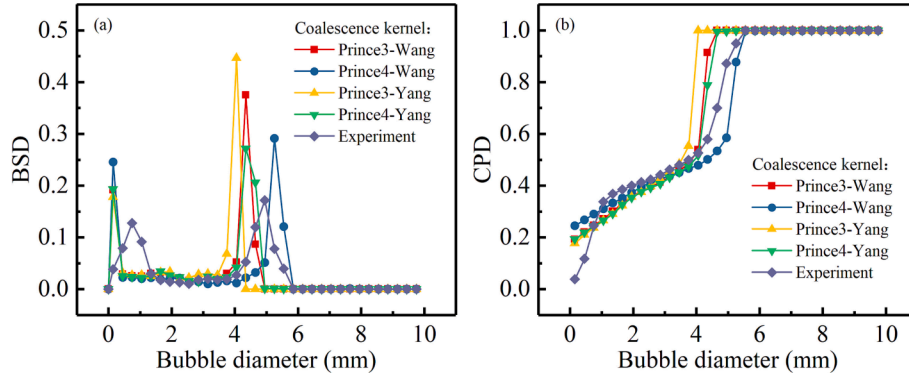


Fig 16. Influence of different coalescence correction coefficients on (a) BSD and (b) CPD at $z/l = 11.5$ under $Re_L = 33000$ and $\alpha_d = 0.05$.

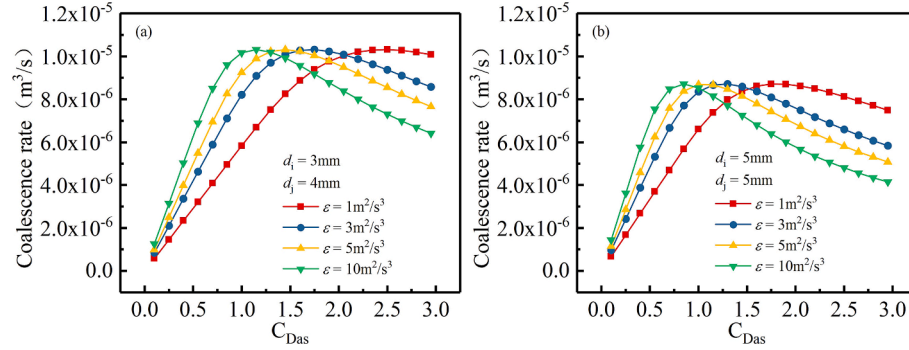


Fig 17. The influence of C_{Das} parameter on bubble coalescence rates.

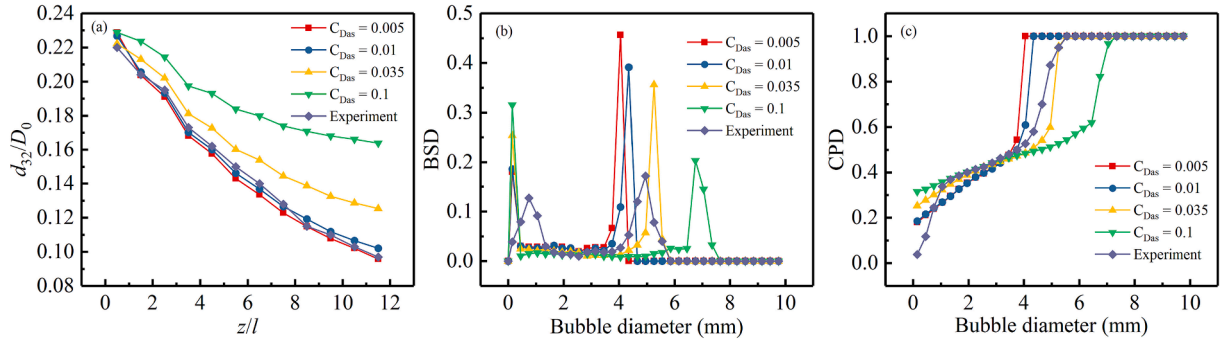


Fig 18. Influence of C_{Das} parameter on: (a) Axial d_{32} , (b) BSD, and (c) CPD at $z/l = 11.5$ using Luo breakup model under $Re_L = 33000$ and $\alpha_d = 0.05$.

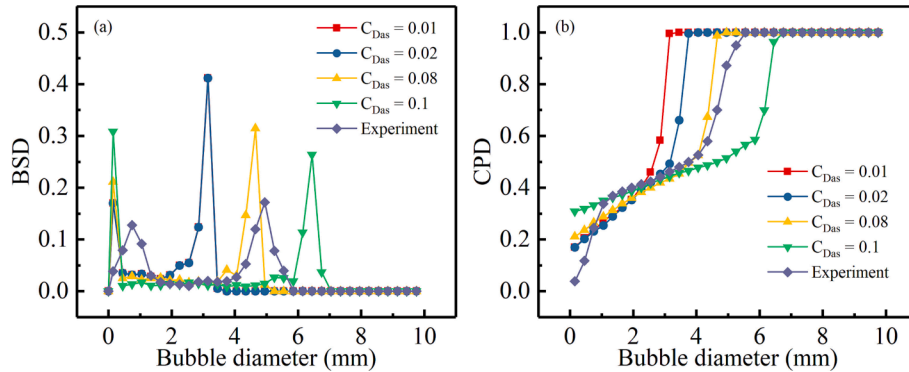


Fig 19. Influence of empirical parameter C_{Das} on the numerical prediction: (a) BSD, (b) CPD at $z/l = 11.5$ when Lehr breakup model is used under $Re_L = 33000$ and $\alpha_d = 0.05$.

4.4. Effects of the empirical constant C_{Das}

Fig. 17 shows the influence of the C_{Das} on the coalescence rate predicted by Das model. As shown in Fig. 17, the bubble coalescence rate calculated by Das model is closely related to the ε and bubbles diameter. The coalescence rate is in uptrend with the rise of empirical constant C_{Das} until it reaches the critical point then decreases. At the same time, the position of the critical point gradually moves to the left with the increase of ε . In addition, the coalescence rate has a gradual increase with the increasement of ε when the C_{Das} is less than 1.

Fig. 18 demonstrates the effect of empirical constant C_{Das} on the calculation results when Luo breakup model is utilized. Fig. 18(a) shows that the bubble coalescence rate in TKSM is significantly overestimated when the C_{Das} is equal to or greater than 0.035. The predicted axial d_{32} can precisely match the experiment when the value of C_{Das} is within the range of 0.005–0.01. Fig. 18(b and c) shows the BSD and CPD at $z/l = 11.5$. Four cases overestimate the number of small bubbles and the increase of parameter C_{Das} also improves the prediction accuracy of the amount of minute bubbles. In addition, the increase of C_{Das} makes the proportion of large bubbles decrease. There is a huge discrepancy in both the simulation results and experimental values when the C_{Das} is 0.035–0.1. The simulation results have better agreement with experimental values when $C_{Das} = 0.01$. Zhang and Luo (2020) found that the most accurate simulation results of local α_G in bubble column reactor were obtained when the $C_{Das} = 0.1$. This shows that the range of C_{Das} is influenced by different gas-liquid flow field structures.

Fig. 19 shows the influence of empirical constant C_{Das} on the simulation results when Lehr breakup model is used. The simulated BSD peak is smaller than the experiment value when C_{Das} is within 0.01–0.08. All in all, the calculated BSD peak increases as the C_{Das} augments. Fig. 19 illustrates that the simulation results have better consistency with the experiment and the MRE between the numerical and experimental CPD is 21.58 % when $C_{Das} = 0.08$. It can be observed from Fig. 18(c and b) that the C_{Das} value corresponding to Lehr breakup model exceeds that of Luo breakup model. This can be attributed to the fact that Lehr breakup model has a higher breakup rate. Therefore, it is necessary to select appropriate empirical parameter C_{Das} to match different breakup models.

4.5. Residence time distribution (RTD) in TKSM

The RTD is an important parameter to investigate the backmixing phenomenon in static mixer. It could be calculated by the tracer concentration. In current research, CFD method is applied to evaluate RTD

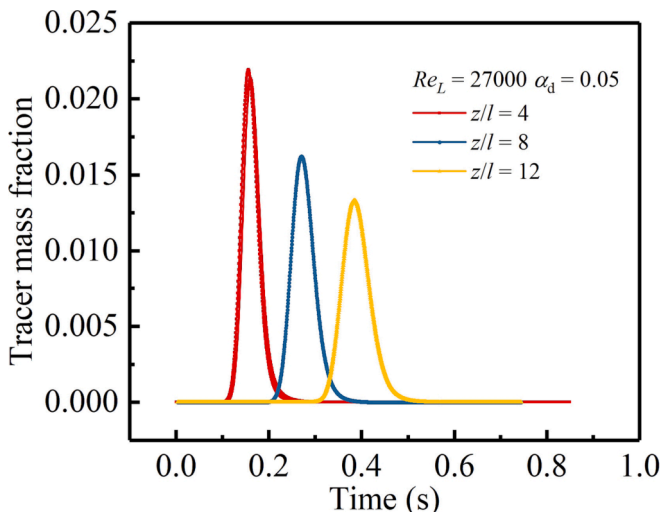


Fig. 20. The liquid RTD at various axial positions.

of gas–liquid two phases in TKSM. Two kinds of tracers which consist of massless particle are separately set as the same physical characteristics as water and air in the continuous phase to ensure that the flow field is not affected.

The mean residence time is defined as:

$$\bar{t}_{r,m} = \frac{\int t c_{r,m} dt}{\int c_{r,m} dt} \quad (27)$$

The RTD variance is calculated via:

$$\sigma_{r,m}^2 = \frac{\int t^2 c_{r,m} dt}{\int c_{r,m} dt} - \bar{t}_{r,m}^2 \quad (28)$$

The normalized variance is calculated via:

$$\sigma_{\theta,m} = \frac{\sigma_{r,m}}{\bar{t}_{r,m}} \quad (29)$$

Fig. 20 demonstrates the liquid RTD at various axial length on the basis of water tracer in TKSM. The peak value of RTD curves decreases with the increase of axial distance, which implies that adding the elements quantities would contribute to a rising trend in the residence time of liquid. The RTD curve of the liquid phase gradually widens when the liquid phase flows from $z/l = 4$ to $z/l = 12$. Strong turbulence is induced from the interaction of water and large sized bubbles, which makes backmixing more likely to occur at the bottom of the static mixer. The turbulence intensity decreases as the large sized bubbles are sheared by the elements and dispersed into minute bubbles, which results in a wider RTD curve.

Fig. 21(a) displays the influence of Re number on the mean residence time (MRT) of liquid phase. The results demonstrate that the MRT of liquid phase decreases as Re increases. The MRT of liquid phase increases by 4.65 % – 25.12 % when Re decreases from 30,000 to 24000. These are mainly due to the fact that the rise of Re improves the liquid velocity in TKSM and reduces the liquid residence time. The MRT of liquid phase increases by 1.24 % – 5.63 % when α_d decreases from 0.08 to 0.02 as depicted in Fig. 21(b). This could be attributable to the further enhancement of BIT effect with the raise of α_d , which promotes the swirling flow. In general, the Re makes a significant difference on the liquid mean residence time compared with the α_d .

Table 6 indicates the residence time of simulation and theoretical plug flow. The simulated liquid residence time is smaller than that of theoretical plug flow. By comparing the liquid residence time obtained by simulation with the plug flow time, the MRE is 3.57 %, 3.65 %, and 3.67 % at $Re = 24000$, 27000, and 30000, respectively. The mixing elements change the fluid flow path, which makes it to flow through an extra distance. It is worth mentioning that the mixing elements decrease the effective flow area and rise the flow velocity, while the increased velocity can offset the losses caused by the additional distance of fluid flow.

$$f = \frac{2D_0 \Delta P}{\rho_c L U_L^2} \quad (30)$$

where D_0 is the inner diameter of tube, ΔP is the pressure drop, L is the length between two middle cross sections of inlet and outlet sections.

The Darcy friction factor (f) in Eq. (30) is adopted to evaluate the pressure drop of experimental and numerical results. Fig. 22(a) depicts the f of experiment and simulation at different U_G and U_L . The f decreases with the increase of U_G at the same U_L . Compared with the experimental f at $U_G = 0.276$ m/s, the deviation of the numerical f at $U_G = 0.276$ m/s is in the range of 6.17 % – 9.76 % and the MRE is within 8.59 %. The dimensionless empirical correlation could be obtained as the following Eq. (31). The application range of this correlation is presented as follows: $0.276 \text{ m/s} \leq U_G \leq 0.774 \text{ m/s}$ and $0.546 \text{ m/s} \leq U_L \leq 0.983 \text{ m/s}$.

$$f = 1.819 U_L^{-1.954} U_G^{-0.0249} \quad (31)$$

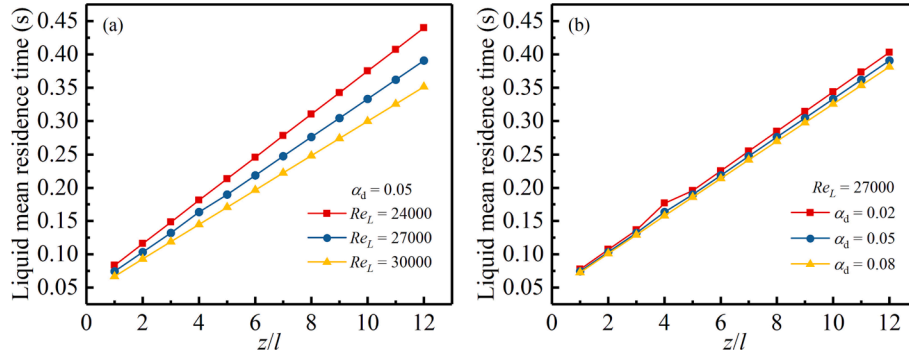


Fig 21. The effects of (a) Re and (b) dispersed phase volume fraction on the MRT of liquid phase.

Table 6

Simulated residence time and theoretical plug flow residence time.

Re	Element number	Plug flow time (s)	Simulated residence time (s)	Deviation (%)
24,000	12	0.4561	0.4398	3.57
27,000	12	0.4055	0.3907	3.65
30,000	12	0.3649	0.3515	3.67

The normalized variance of liquid phase is shown in Fig. 23. The liquid phase normalized variance reaches its maximum value at the first element and decreases as z/l increases. The liquid and gas phases entering from the bottom of the mixer collide strongly, which further improves the backmixing efficiency. At the same time, a larger Re means a stronger energy dissipation rate, which leads to more Kolmogorov scale eddies to further intensify the radial secondary flow. These make the turbulence of liquid phase more intense and improve liquid phase normalized variance.

Fig. 24 shows the effect of Re and α_d on the MRT of gas phase. The turbulence and complexity of the flow field increases as Re varies from 24,000 to 30,000, which leads to the enhancement of backmixing as described in Fig. 23. The coalescence rate could be enhanced by backmixing in the first five elements. Therefore, the MRT of gas phase decreases with the increase of Re as shown in Fig. 24(a). On the contrary, the MRT of gas phase rises with the increase of Re in the range of $z/l = 5-12$. This may be because more bubbles are further broken by enhanced radial secondary eddies. Generally speaking, the MRT of the fluid declines with the enhancement of flow rate as shown in Fig. 24(b). When the α_d is 0.02, 0.05, and 0.08, the corresponding gas phase velocity at $Re = 27000$ is 0.186 m/s, 0.4797 m/s and 0.792 m/s, respectively. The flow rate of gas phase rises as α_d increases, thereby reducing the MRT of gas phase. Overall, the MRT of gas phase increases because of the rise of element number. The MRT of gas phase increases by 269.41 % – 332.16 % when the fluid flows from the first element to

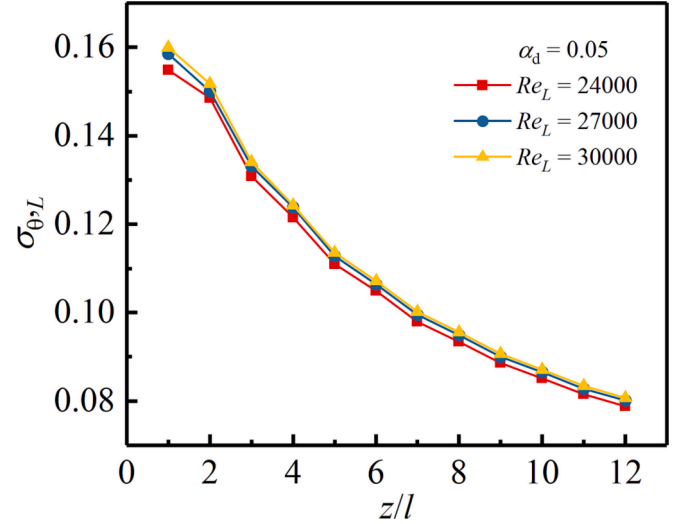


Fig 23. Normalized variance of liquid phase at various axial positions.

the twelfth element. This phenomenon is mainly because of the shear effect of the mixing element on initial large sized bubbles, which induces them to gradually break into small bubble swarms. In addition, it is worth mentioning that the influence of α_d on gas mean residence time is weakened as the α_d increases.

The axial normalized variance of gas phase depicted in Fig. 25 reaches its peak at the first element then declines as flow length rises. The trend indicates that the higher gas phase velocity increases the coalescence rate and the quantities of large sized bubbles, which is different from vortex unit (Chen et al., 2022). Meanwhile, the higher rising speed of large sized bubbles increases the mixing efficiency. In addition, the rise of flow rate of gas phase may lead to a low pressure drop, thereby

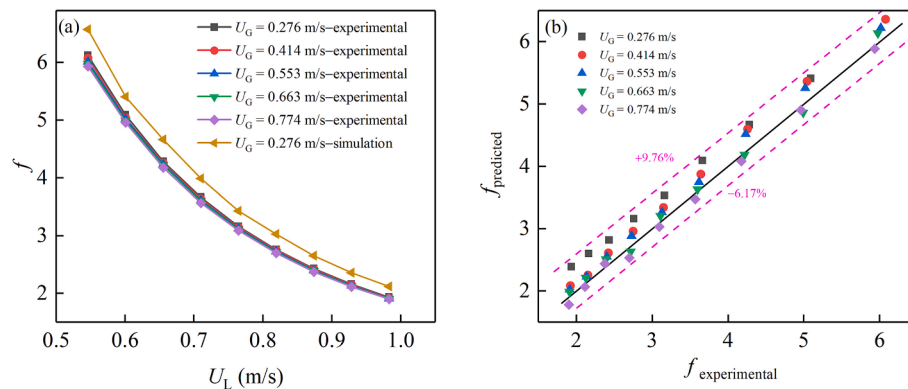


Fig 22. The comparison between darcy fraction factors of experiment and numerical simulation.

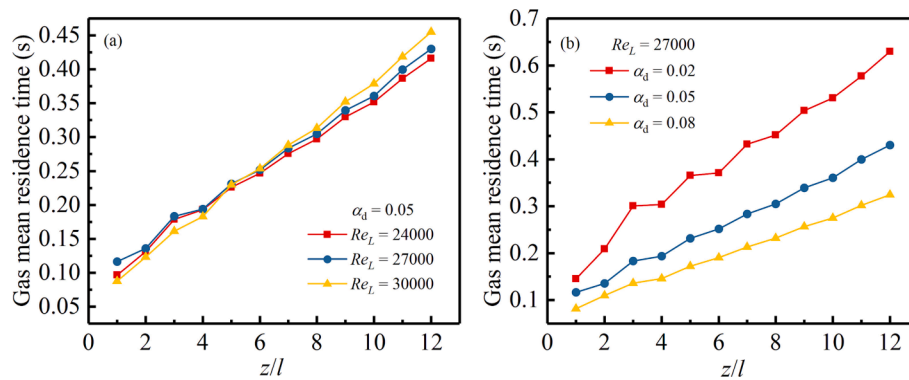


Fig 24. The effects of (a) Re and (b) dispersed phase volume fraction on the MRT of gas phase.

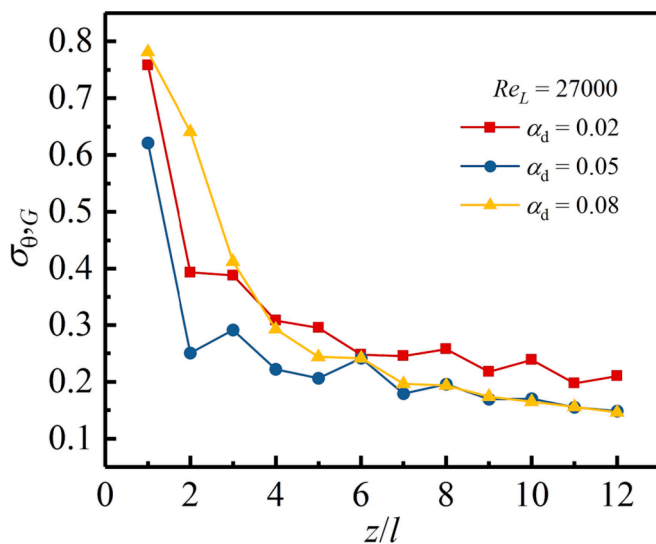


Fig 25. Normalized variance of gas phase at different axial positions.

reducing the gas normalized variance. As is well known, different mixer structures have different effects on the pressure drop, so it is essential to choose an appropriate static mixer to improve mixing performance.

5. Conclusions

In current work, the influences of breakup and coalescence kernels on the numerical predictions in TKSM are studied. In addition, the impact of bubble swarm effect on bubble dynamic behavior in TKSM is investigated. Some conclusions are obtained.

- (1) The Brucato model overestimates the radial α_G at the center and underestimates the radial α_G near wall surface. The mean relative error among the calculated results of Schiller-Naumann model, Tomiyama model, Grace model and experiment also increases when the U_G increases. The modified model coupled with correction factors can effectively improve prediction accuracy.
- (2) Turbulence and Liao coalescence model underestimate the bubbles coalescence rate in TKSM, while Luo coalescence model overestimates the bubbles coalescence rate in TKSM. In comparison with the axial d_{32} obtained by Lehr-Prince model, the prediction accuracy of Luo-Prince model is improved by 6.84 %. Overall, Luo-Prince model has the best prediction results.
- (3) Turbulent fluctuation is the most important factor leading to the coalescence behavior of bubbles. Secondly, it is viscous shear mechanism. The contribution of wake entrainment and buoyancy driven to bubble coalescence is negligible. Additionally, the

prediction precision of CPD is enhanced by 17.26 % when the coalescence correctors obtained by EMMS method is utilized.

- (4) The accurate BSD prediction has been achieved. The range of C_{Das} value is 0.01–0.035 when Luo-Das model is used and the range of C_{Das} value is 0.08–0.1 when Lehr-Das model is used.
- (5) The backmixing degree of the gas phase exceeds that of the liquid phase because of bubble coalescence and breakup behavior in TKSM. The mean residence time of the continuous phase is smaller than that of the dispersed phase.

For the improvement of the prediction results of bubble dynamics performance in static mixers, the contribution of different bubble breakage mechanisms to bubble breakup behavior would be studied in the further research. For example, the bubble deformation, the transport of large-scale eddies and the effect of full energy spectrum function should be considered. Furthermore, it is hoped that future research would focus on the influence of mixer structures on pressure drop and the gas phase variance.

CRedit authorship contribution statement

Yanfang Yu: Conceptualization, Investigation, Methodology, Supervision, Writing – original draft. **Jiawei Zhang:** Methodology, Software. **Yu Li:** Investigation, Software, Visualization. **Kexin Xiang:** Investigation, Software, Visualization. **Huibo Meng:** Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was supported by the following funds: the Key Scientific Research Project of Education Department of Liaoning Province (LJKZ0429, LJKMZ20220773), Qingdao Natural Science Foundation (23-2-1-236-zyyd-jch), Shenyang Young and Middle-aged Science and Technology Innovation Talent Support Program (RC200032), National Natural Science Foundation of China (21476142), Distinguished Professor of Liaoning Province (LCH[2018]35), Natural Science Foundation of Liaoning Province (2022-MS-290), and Talent Introduction Research Fund of China University of Petroleum (East China) (R20220113). We

thank the referees for their enlightening remarks which helped us improve the manuscript.

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